THE UNIVERSITY OF HONG KONG

DEPARTMENT OF MATHEMATICS

MATH2012: Fundamental Concepts of Mathematics

May 24, 2018 2:30pm - 5:00pm

Only approved calculators as announced by the Examinations Secretary can be used in this examination. It is candidates' responsibility to ensure that their calculator operates satisfactorily, and candidates must record the name and type of the calculator used on the front page of the examination script.

Answer ALL EIGHT questions

Note: You should always give precise and adequate explanations to support your conclusions. Clarity of presentation of your argument counts. So **think carefully before you write.**

1. (*15 points*) For each of the questions below there is only ONE correct answer. Write your answers in the **ANSWER BOOK** provided. No steps/explanations are required.

(1) Which of the following statements is/are TRUE for all subsets A, B, C of a universal set U?

I. If $A \subseteq B$, then $A \backslash C \subseteq B \backslash C$;
II. $P(A \cup B) = P(A) \cup P(B)$;
III. $P(A \cap B) = P(A) \cap P(B)$;
IV. If $|A \cup C| = |B \cup C|$, then $|A| = |B|$.

A. I only; B. I and III only; C. II and III only; D. I, III and IV only.

(2) For two statements P and Q, which of the following statements is/are logically equivalent to the statement $P \to Q$?

I. $(\sim P) \vee Q$;
II. $(\sim Q) \to (\sim P)$;
III. $(\sim (\sim Q)) \vee (\sim P)$.

A. I only; B. II only; C. I and II only; D. I, II and III.

(3) For $A \subseteq \mathbb{R}$, which of the following is the negation of the statement that "There exists $M > 0$ such that $a > M$ for all $a \in A$"?

I. $\exists M > 0$ such that $a \leq M$ for all $a \in A$;
II. $\forall M > 0, \exists a \in A$ such that $a > M$;
III. $\forall M > 0, \exists a \in A$ such that $a \leq M$.

A. I only; B. II only; C. III only; D. None of these.

(4) Let $\{a_n\}_{n=1}^{+\infty}$ be a sequence of real numbers and let $a \in \mathbb{R}$. Which of the following is/are, or equivalent to, the definition of "$\lim\limits_{n \to +\infty} a_n = a$"?

I. $\forall \varepsilon > 0$, $\exists N \in \mathbb{N}$ such that $|a_n - a| < \varepsilon$ for all $n \in \mathbb{N}$ and $n > N$;
II. $\forall \varepsilon > 0$, $\exists N \in \mathbb{N}$ such that $|a_n - a| < \varepsilon$ for all $n \in \mathbb{N}$ and $n \geq N$;
III. $\forall \varepsilon > 0$, $\exists N \in \mathbb{N}$ such that $|a_n - a| \leq \varepsilon$ for all $n \in \mathbb{N}$ and $n > N$;
IV. $\forall \varepsilon > 0$, $\exists N \in \mathbb{N}$ such that $|a_n - a| \leq \varepsilon$ for all $n \in \mathbb{N}$ and $n \geq N$.

A. I only; B. I and II only; C. I, II, and III only ; D. All of these.

(5) Let $\{a_n\}_{n=1}^{+\infty}$ be a sequence of real numbers. Which of the following is/are correct?

I. If there exists $M > 0$ such that $|a_n| < M$ for all $n \in \mathbb{N}$, then the sequence $\{a_n\}_{n=1}^{+\infty}$ has a limit;

II. If the series $\sum_{n=1}^{+\infty} a_n$ converges, then $\lim\limits_{n\to+\infty} a_n = 0$;

III. If $\lim\limits_{n\to+\infty} a_n = 0$, then the series $\sum_{n=1}^{+\infty} a_n$ converges.

A. II only; B. I only; C. II and III only; D. All of these.

2. (*10 points*) Prove or disprove the following statements:

1) For any positive rational numbers a and b, the number a^b is again rational.

2) There exist positive irrational numbers a and b such that a^b is rational.

3. (*10 points*) Find all integers x and y satisfying

$$\begin{cases} x + y^2 \equiv 1 \bmod (3), \\ x - y^2 \equiv 2 \bmod (3). \end{cases}$$

4. (*10 points*) Let $A = \{(m, n) : m, n \in \mathbb{Z}, n \neq 0\}$ and define $f : A \to \mathbb{Q}$ by

$$f(m, n) = \frac{m+n}{n}, \quad (m, n) \in A.$$

Is f injective? If f surjective? Find also $f^{-1}(\mathbb{Z})$. Explain your answers.

5. (*10 points*) Let A be the set of all maps from $\mathbb{N}$ to $\mathbb{N}$. Prove that A is uncountable.

6. (*15 points*) Consider the relation $\sim$ on $\mathbb{R}$ defined by

$$x \sim y \quad \text{if and only if} \quad x - y \in \mathbb{Z}.$$

1) Prove that this relation is an equivalence relation;

2) Describe the equivalence class $[\sqrt{2}]$ of $\sqrt{2}$. Show that every equivalence class of $\sim$ is a countable subset of $\mathbb{R}$.

3) Let $\mathbb{R}/\sim$ be the set of all equivalence classes of $\sim$ in $\mathbb{R}$. Show that the map

$$f : \mathbb{R}/\sim \longrightarrow \mathbb{R}^2, \quad f([x]) = (\cos(2\pi x), \sin(2\pi x)),$$

is well-defined. Is f injective? What is the image of f?

7. (*10 points*) For an integer $n \geq 1$, let F_n be the rational number given by the continuous fraction

$$F_n = 2 + \cfrac{1}{2 + \cfrac{1}{2 + \cfrac{1}{\cdots + \cfrac{1}{2 + \cfrac{1}{2}}}}},$$

where the number 2 appears n times. For example, $F_1 = 2$ and $F_2 = 2 + \frac{1}{2} = \frac{5}{2}$. Let $K_0, K_1, K_2, K_3, \ldots$ be the sequences of real numbers given by $K_0 = 1, K_1 = 2$, and

$$\frac{K_n}{K_{n-1}} = F_n, \quad \text{for} \quad n \geq 2.$$

Show that for every $n \geq 1$, K_n is a positive integer and that K_n and K_{n-1} are relatively prime.

8. (*20 points*) Consider the sequence

$$a_n = \sqrt{2 + \sqrt{2 + \sqrt{2 + \cdots + \sqrt{2}}}}, \quad n \in \mathbb{N},$$

where for $n \in \mathbb{N}$, the number 2 appears n times in the expression for a_n.

1) Prove that $\lim_{n \to +\infty} a_n$ exists and find the limit.

2) Let $b_n = 2 - a_n$ for $n \in \mathbb{N}$. Show that $\sum_{n=1}^{\infty} b_n$ is convergent.

******* **END OF PAPER** *******